Hana Petrović

Machine Learning Research Scientist (PhD) · Zurich, Switzerland · hana.petrovic@example.com

Summary

Research scientist working on language models for long documents: retrieval, evaluation and efficient fine-tuning. PhD in natural language processing with a minor in computer vision. I enjoy turning a paper idea into a system that real users can rely on, and measuring honestly whether it works.

Experience

Research Scientist — Lumina AI · Sep 2022 - Present
- Train and fine-tune transformer models in PyTorch (LoRA and full fine-tuning), with distributed training on 64 GPUs.
- Built retrieval-augmented generation (RAG) for legal documents on a vector database; recall@10 up from 0.61 to 0.83.
- Designed the LLM evaluation suite (faithfulness, citation accuracy, human ratings) used for every release.
- Experiment tracking in Weights & Biases; supervise two research interns.

Doctoral Researcher, NLP group — ETH Zurich · Sep 2018 - Aug 2022
- Thesis on cross-document reasoning; models in PyTorch, analysis in Python with NumPy, scikit-learn and Jupyter.
- Side project in computer vision: layout detection for scanned contracts.
- Teaching assistant for Introduction to Machine Learning and Deep Learning (three semesters).
- Statistical testing and careful experimental design across all published work.

Education

ETH Zurich, Ph.D. Computer Science (natural language processing) (2018 - 2022)

University of Belgrade, M.Sc. Software Engineering (2016 - 2018)

Publications

Petrović H., Meier L. Reasoning across documents with sparse memory. ACL 2022.
Petrović H., Rossi F., Meier L. When retrieval hurts: an analysis of long-document QA. EMNLP 2021.
Petrović H., Meier L. Contract layout as a signal for clause classification. ICDAR 2021.
Rossi F., Petrović H. A benchmark for citation faithfulness in generated answers. NAACL 2023.
Petrović H., Chen Y., Rossi F. Efficient fine-tuning for long contexts. Findings of ACL 2024.
Meier L., Petrović H. Teaching assistants for ML courses: lessons from three years. EAAI 2022.

Talks

Evaluating RAG systems you can trust. Swiss Machine Learning Days 2024 (invited talk).
Long documents, short attention. PyData Zurich 2023.

Awards

ETH Medal for an outstanding doctoral thesis (2023).
Best paper honourable mention, EMNLP 2021.